Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/15/2015
PI Name:	Haswell, Elizabeth Ph.D.		
Project Title:	Mechanosensitive Ion Channels in Plants: Genetic, Computational and Systems-Level Approaches to Understanding their Proposed Role in Gravity Perception		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
PI Email:	ehaswell@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314-935-9223
Organization Name:	Washington University in St. Louis		
PI Address 1:	Department of Biology		
PI Address 2:	1 Brookings Dr, Box 1137		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130-4862	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	08/01/2013	**End Date:**	07/31/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Levine, Howard	**Contact Phone:**	321-861-3502
Contact Email:	howard.g.levine@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2017 per NSSC information (Ed., 5/5/16)		
Key Personnel Changes/Previous PI:	May 2015 report: Maggie Wilson, PhD 50% effort; Matt Mixdorf, technician 100% effort; Samantha Embrick, undergraduate work-study, approximately 3 hours per week.		
COI Name (Institution):	Spalding, Edgar Ph.D. (University of Wisconsin-Madison)		
Grant/Contract No.:	NNX13AM55G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	How plants respond to gravity is a fundamental biological question with important implications for space flight. Here we propose ground-based research designed to lead to spaceflight, in the area of plant biology. Our experimental approach focuses on several families of putative mechanoreceptors known as mechanosensitive (MS) ion channels. MS ion channels have been proposed to play a role in the perception of gravity in plants for many years, a hypothesis that is supported by several lines of evidence, but one that has not yet been systematically tested by molecular genetics. The research plan utilizes contemporary methodologies for the study of the model plant Arabidopsis thaliana to interrogate the biological function of plant MS ion channels and builds upon previous NASA-funded research into the response of plants to altered gravity environments. We will determine the role played by MS ion channels in mediating gravity signal transduction, their putative roles in adaptation to microgravity, and the molecular mechanisms by which the model MS ion channel MSL10 activates stress-response pathways in plant cells. We expect that these activities will have the following outcomes: 1) Test the long-standing hypothesis that MS ion channels function in gravity perception; 2) Contribute to our basic knowledge regarding how plants exploit MS ion channels to sense and respond to mechanical stress; 3) Provide information that could be used on Earth to improve crop productivity; and 4) Establish a foundation of knowledge required to generate more mechanistic and sophisticated hypotheses about plant gravity perception that can be tested in spaceflight using the European Modular Cultivation System and/or the Advanced Biological Research System (ABRS) Green Fluorescent Protein (GFP) Imaging system. (These proposed studies directly address the objectives of		

	the solicitation, as they aim to answer basic questions about how plants respond to changes in gravity and should lead to hypotheses that can be tested on future spaceflights.)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Identifying the molecular mechanisms by which MS ion channels function and evaluating their roles in the perception and response to altered gravity conditions will help fill a large gap in our knowledge regarding the molecules involved in gravity and mechanoperception. Gaining this knowledge has the potential to achieve two long-term goals: 1) to mitigate the effects of mechanical stress on crop plants on Earth, and 2) to counteract the effects of altered gravity environments on plants grown in space. These experiments will address the high priority goal of "understanding the basic mechanisms of plant response" to changes in gravity described in the Decadal survey of 2011.
Task Progress:	Aim 1: Test the Hypothesis that MS Ion Channels Play a Role in Root Gravitropism Using a Machine Vision and Computation Approach Introduction: Mechanosensitive (MS) ion channels have been implicated in a variety of steps in gravity signaling, but are most often implicated in the initial gravity response. MS ion channels have been proposed to sense ER (endoplasmic reticulum) membrane tension as a result of the amyloplasts sedimentation; the plasma membrane tension resulting from the uneven pressure of the tonoplast at the bottom of the cell; or the action of amyloplasts on actin microfilaments tethered to the plasma membrane. There are at least three potential barriers to the identification of MS ion channels in forward genetic screens for genes involved in gravity response: 1) redundancy of MS ion channel gene function; 2) redundancy of gravity response pathways involving MS ion channels; and 3) subtle effects of MS ion channel function on the gravity response. We aim to test the long-held hypothesis that MS ion channels mediate the perception of changes in gravity vector by using a reverse genetics approach and a highly sensitive Machine Vision and Computation assay for defects in gravity perception. Progress: A. We produced three new higher order mutants lacking functional mechanosensitive ion channels and/or components of known gravity signaling pathways. To determine the role of MS channels in gravity response, we are combining mutations that produce lesions in multiple channels and/or signaling pathways. We have made msl9 msl10 arl2 triple mutants, msl4 msl5 msl6 msl9 msl10 arl2 hextuple mutants; and msl4 msl5 msl6 msl9 msl10 pgm-1 hextuple mutant lines. These lines have been sent to Edgar Spalding at U. Wisconsin (a co-PI on this grant) for root gravitropic analysis. We are also working on removing a background mutation from the mca1 mca2 double mutant, troubleshooting genotyping the mca1 allele, and creating a knock-out of the AtPiezo MS ion channel gene using new CRISPR/Cas technology, as it appears to be essential. B. We initiated machine imaging of root gravitropism of higher order mechanosensitive ion channel mutants. Previously generated lines were sent to Edgar Spalding at U. Wisconsin (a co-PI on this grant) for root gravitropic analysis. We found that a lesion in the MSL ion channel gene MSL4 causes seedlings to overshoot the maximum angle of bending (or have a higher "swing rate") than wild type. Furthermore, the msl2 msl3 double mutant exhibits slow gravity response, though to a more subtle degree than previously established gravity mutants. C. We further characterized the gravity response defect of the msl2 msl3 double mutant. The msl2 msl3 double mutant lacks MS channels in the plastid envelope. This mutant shows overall defects in orientation when grown vertically on solid media. We have been trying to determine if this defect in root gravitropism is due to defects in amyloplast development, amyloplast sedimentation, or other effects. Recent results suggest that the defect may be attributable to a loss of columella cell identity in these mutant lines. The statoliths disappear or appear disorganized compared to those in the wild type. These mutants, along with appropriate controls, are growing or will be grown for shoot gravitropism analysis in collaboration with Darron Luesse at SIUE (Southern Illinois University - Edwardsville). Aim 2: Test the Hypothesis that MSL and Other MS Ion Channels Are Involved in the Perception and/or Adaptation to Weightlessness. Progress: We do not have progress to report on this aim. Aim 3: Determine the Molecular Mechanism of MSL10-Induced Reactive Oxygen Species (ROS) Signaling Introduction: MS ion channels are implicated in the perception and/or response to changes in gravity conditions. The experiments in this aim are designed to gain a more molecular and mechanistic handle on the regulation and signaling behavior of MS ion channels, building on our recent observations that overexpressing the MscS-Like MS ion channel MSL10 from A. thaliana leads to ROS-associated cell death. Understanding how MSL10 is regulated and how its activation leads to ROS-related stress signaling may be important for mitigating the effects of mechanical stress on crop plants or plants in the space environment. We previously reported that MSL10 induces cell death signaling in a phosphorylation-dependent manner and that the N-terminus of MSL10 alone can provide this activity, independent of the ion channel function of the channel. We continue to suspect that 14-3-3 phospho-binding proteins (also called GRFs) may play a role in regulating MSL10 activity and are working on testing this idea. Progress: A. We explored a number of approaches to test for interactions between 14-3-3 proteins and MS ion channels. We tried the yeast two-hybrid but found that 14-3-3 proteins routinely auto-activated in that system. We then cloned and purified recombinant 14-3-3 proteins along with the soluble N-termini of MSL8, MSL9, and MSL10 produced in E. coli. Experiments are underway to assess direct interactions between these proteins. B. We developed an assay for ROS production in response to osmotic shock. Wild-type and mutant seedlings were grown in triplicate for 8 days in 125 ml flasks containing 25 ml of 1X MS liquid media either with or without supplemental 50 mM NaCl for osmotic stress. Prior to the assay, the media in each flask was replaced with fresh, and the plants were allowed to equilibrate for 4 hours. For osmotic shock, seedlings were transferred to a multi-well plate and the samples were either subjected to up-shock (transferred from 1X MS to 1X MS with 50 mM NaCl) or down-shock (transferred from 1X MS with 50 mM NaCl to 1X MS). We hope to use this assay to to identify the MSL10 kinase and phosphatase, and develop a time course of gene expression response to osmotic shock and ROS production. C. We pursued the effect and the source of ROS overproduction in msl2 msl3 mutants. We observed that msl2 msl3 double mutants develop large, bulbous callus-like growths on the shoot apex after growth on solid media, as shown below. These growths are associated with the extreme over-production of ROS. We are pursuing a number of lines of experimentation to provide support for our current hypothesis that a signaling pathway associated with the ratio of the phytohormone cytokinin to the phytohormone auxin induces ROS production and callus formation in this mutant plant. msl2 msl3 double mutants are indeed high overproducers of cytokinin, and auxin treatment can suppress callus production. Summary In conclusion, we have made good progress on both the first and the third aim over the past year and are set up to collect a great deal more data and publish several papers in the coming year.
Bibliography Type:	Description: (Last Updated: 08/25/2020)
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